



6278 - BonFIRE: Modeling Galaxy Formation in the Early Universe

Cycle: 3, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Results from JWST on high-redshift galaxy formation appear to be in tension with theoretical predictions, but current simulations lack the necessary resolution, volume coverage, and detailed physics to accurately model the high-redshift galaxy population. We propose to use a new large-volume hydrodynamic simulation, BonFIRE, to bridge this gap and make robust predictions for early galaxy formation that are ideal for comparison with JWST observations. BonFIRE will simultaneously leverage high resolution and a large volume to make robust predictions for the formation histories of early galaxies and the statistical evolution of the galaxy population at $z=8-20$. The simulation will allow us to address crucial questions about the

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onset of galaxy formation, the conditions for early galaxy formation, the number of early galaxies, the role of galaxies in reionizing the Universe, and the emergence of disk morphology. Using BonFIRE, we will make predictions for the ultraviolet luminosity function, the number density of massive galaxies, and the physical mechanisms driving the growth of early galaxies. We plan to re-run the simulation with additional physics models such as AGN feedback and Population III stars to test their effects on the high-redshift galaxy population. We will also generate and publicly release mock observations of the simulated galaxies for direct comparison to observations from JWST. This work will provide insight into the conditions for galaxy formation in the early Universe and set a critical theoretical benchmark for observations of high-redshift galaxies.

OBSERVING DESCRIPTION